Exponential Functions and Their Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating and describing

Let $f(x) = 3 \cdot 2^x$. Evaluate:

a $f(0) = 3$ **b** $f(3) = 24$ **c** $f(-2) = \frac{3}{4}$

(a) $= 3$; (b) $= 24$; (c) $= \frac{3}{4}$

2 [Foundational] • Target: Evaluating and describing

For $y = 5^x$, state:

a the domain All real numbers.

b the range $y > 0$.

c the horizontal asymptote $y = 0$.

d the y-intercept $(0, 1)$.

(a) All real numbers.; (b) $y > 0$.; (c) $y = 0$.; (d) $(0, 1)$.

3 [Foundational] • Target: Evaluating and describing

Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g.

Shift right 3 units and up 1 unit. Horizontal asymptote: $y = 1$.

4 [Foundational] • Target: Identifying from graphs and applications

The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.

$y = 2^x$.

5 [Foundational] • Target: Identifying from graphs and applications

An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%: $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.

$A(5) = 1000e^{0.3} \approx 1350$ riyals.

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6 [Foundational] • Target: Identifying from graphs and applications

Determine whether each function represents growth or decay:

a $y = 4(1.08)^x$ Growth.

b $y = 7(0.85)^x$ Decay.

c $y = 2e^{-0.3x}$ Decay.

d $y = \left(\frac{5}{4}\right)^x$ Growth.

(a) Growth.; (b) Decay.; (c) Decay.; (d) Growth.

7 [Foundational] • Target: Compound interest

2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.

a Find the value after 3 years, to the nearest riyal. $A = 2000\left(1 + \frac{0.05}{12}\right)^{36} \approx 2000(1.1616) \approx 2323$ riyals.

b Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal. $A = 2000e^{0.15} \approx 2000(1.1618) \approx 2324$ riyals.

(a) $A = 2000\left(1 + \frac{0.05}{12}\right)^{36} \approx 2000(1.1616) \approx 2323$ riyals.; (b) $A = 2000e^{0.15} \approx 2000(1.1618) \approx 2324$ riyals.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Compound interest

How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

$P = \frac{10000}{(1.04)^8} \approx \frac{10000}{1.3686} \approx 7308$ riyals.

9 [Intermediate] • Target: Comparing exponential growth rates

Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.

4%: $500(1.04)^{10} \approx 740$ riyals. 7%: $500(1.07)^{10} \approx 984$ riyals. Difference ≈ 244 riyals.

10 [Intermediate] • Target: Comparing exponential growth rates

Rank the following functions from slowest to fastest growth as $x \to \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.

Any exponential with base > 1 eventually outgrows any polynomial, no matter how slowly it starts. So even though 1.01^x grows very slowly at first, it eventually overtakes x^{10}. Long-run ranking (slowest to fastest): $x^{10} < 1.01^x < 2^x$.

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11 [Intermediate] • Target: Comparing growth and decay graphically

The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.

The two curves are mirror images of each other across the y-axis: $\left(\frac{1}{2}\right)^x = 2^{-x}$, so reflecting $y = 2^x$ over the y-axis (replacing x with $-x$) produces the decay curve.

12 [Intermediate] • Target: The natural exponential base e

Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.

The number e is the unique base for which the exponential function $y = b^x$ has a tangent line slope of EXACTLY 1 at $x = 0$ — i.e., the function's instantaneous rate of change equals its own value at every point, a property no other base shares.

13 [Intermediate] • Target: The natural exponential base e

Evaluate e^0, e^1 (to 3 decimal places), and simplify $e^{\ln 5}$.

$e^0 = 1$. $e^1 \approx 2.718$. $e^{\ln 5} = 5$ (since e^x and $\ln x$ are inverse functions).

14 [Intermediate] • Target: Piecewise-exponential applications

A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).

$V(5) = 15000(0.88)^5 \approx 15000(0.5277) \approx 7916$ riyals. Setting $15000(0.88)^t < 5000$: $(0.88)^t < 0.333$. Testing values: at $t = 9$, $(0.88)^9 \approx 0.316 < 0.333$; at $t = 8$, $(0.88)^8 \approx 0.360 > 0.333$. First drops below 5000 riyals in year 9.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

State the horizontal asymptote and range of each transformed exponential function:

a $y = 3^x + 4$ Asymptote $y = 4$; range $y > 4$.

b $y = -2^x$ Asymptote $y = 0$; range $y < 0$ (reflected below the axis).

c $y = 5^x - 3$ Asymptote $y = -3$; range $y > -3$.

(a) Asymptote $y = 4$; range $y > 4$.; (b) Asymptote $y = 0$; range $y < 0$ (reflected below the axis).; (c) Asymptote $y = -3$; range $y > -3$.

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16 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Write the equation of the function obtained by taking $y = e^x$, reflecting it across the y-axis, then shifting it down 2 units.

Reflecting across the y-axis replaces x with $-x$: $y = e^{-x}$. Shifting down 2: $y = e^{-x} - 2$.

17 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x + 1) = 3f(x)$.

$f(-1) = \frac{1}{3}$, $f(0) = 1$, $f(1) = 3$, $f(2) = 9$, $f(3) = 27$. Check: $f(0) = 3f(-1) = 3(1/3) = 1$ ✓; $f(1) = 3f(0) = 3$ ✓; $f(2) = 3f(1) = 9$ ✓; $f(3) = 3f(2) = 27$ ✓ — each step multiplies by the base 3, confirming the defining property of exponential growth.

18 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

Write $y = \frac{3x-1}{2x+5}$. The domain of f is $x \neq -2.5$. Its horizontal asymptote is $y = 3/2$, so the range of f is $y \neq 1.5$. Swap x and y and solve for y: $x = \frac{3y-1}{2y+5} \Rightarrow x(2y + 5) = 3y - 1$. $2xy + 5x = 3y - 1 \Rightarrow 2xy - 3y = -5x - 1$. $y(2x - 3) = -(5x + 1) \Rightarrow y = \frac{-5x-1}{2x-3} = \frac{5x+1}{3-2x}$. Inverse function: $f^{-1}(x) = \frac{5x+1}{3-2x}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (the range of f). Range of f^{-1}.

19 [Challenging] • Target: Evaluating and describing • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve the exponential equation $3^{2x-1} = 5^{x+2}$ for x in terms of natural logarithms.

Step 1: Take the natural logarithm of both sides: $\ln(3^{2x-1}) = \ln(5^{x+2}) \Rightarrow (2x - 1)\ln(3) = (x + 2)\ln(5)$. Step 2: Expand and group all terms containing x: $2x\ln(3) - \ln(3) = x\ln(5) + 2\ln(5) \Rightarrow x[2\ln(3) - \ln(5)] = 2\ln(5) + \ln(3)$. Step 3: Solve for x: $x = \frac{\mathbf{2\ln(5) + \ln(3)}}{\mathbf{2\ln(3) - \ln(5)}} = \frac{\ln(\mathbf{75})}{\ln(\mathbf{9/5})} \approx 7.37$.

20 [Challenging] • Target: Identifying from graphs and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t) = 250e^{-0.048t}$, where t is measured in days. Determine the exact daily multiplicative decay factor b in the form $M(t) = 250b^t$, and calculate the daily percentage loss rounded to two decimal places.

Multiplicative Base: Rewrite $e^{-0.048t} = (e^{-0.048})^t$: $b = e^{-0.048} \approx 0.953134$. The exact base is $b = e^{-0.048}$ (or approximately 0.9531). Daily Percentage Loss: The percentage decay rate per day is $1 - b$: $1 - 0.953134 = 0.046866 \Rightarrow 4.69\%$. The daily percentage loss is 4.69% (distinguished from the continuous rate of 4.80%).

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating and describing

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$ ✓

b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

Applying Exponential & Logarithmic Functions principles, As $n \to \infty$: $A = Pe^{rt}$. Therefore, the verified correct option is $A = Pe^{rt}$.

22 [Foundational] • Target: Identifying from graphs and applications

The graph of $y = -2f(x + 1) - 3$ applies which transformations to $y = f(x)$?

a. Left 1, stretch 2, down 3 (no reflect)

b. Right 1, reflect, stretch 2, up 3

c. Right 1, stretch 2, down 3

d. **Left 1, reflect over x-axis, vertical stretch 2, down 3** ✓

Inside: $x + 1 \to$left 1. Outside: $-2 \cdot \to$reflect+stretch×2, $-3 \to$down 3.

23 [Foundational] • Target: Compound interest

Express $\log_3 7$ using the change of base formula with natural log.

a. $\frac{\ln 7}{\ln 3}$ ✓

b. $7 \ln 3$

c. $\frac{\ln 3}{\ln 7}$

d. $\ln 7 - \ln 3$

Change of base: $\log_b x = \frac{\ln x}{\ln b}$.

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24 [Foundational] • Target: Comparing exponential growth rates

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

a. 1 ✓

b. -1

c. 2

d. -2

The function is given by $f(x) = ax^2 + bx - 5$. Since the graph passes through $(-2, 3)$, we have $f(-2) = 3$: a(-2)^2 + b(-2) - 5 = 3 \\ 4a - 2b - 5 = 3 \\ 4a - 2b = 8 \\ 2a - b = 4 \quad (1) Since the graph passes through $(1, -3)$, we have $f(1) = -3$: a(1)^2 + b(1) - 5 = -3 \\ a + b - 5 = -3 \\ a + b = 2 \quad (2) Now we have a system of two linear equations with two variables: 1) $2a - b = 4$ 2) $a + b = 2$ Adding equation (1) and equation (2): (2a - b) + (a + b) = 4 + 2 \\ 3a = 6 \\ a = 2 Substituting.

25 [Foundational] • Target: Comparing growth and decay graphically

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

a. **17** ✓

b. 13

c. 20

d. 16

To find the composite function value $(f \circ g)(2)$, we evaluate the inner function first, and then apply the outer function: 1. Evaluate $g(2)$: $g(2) = 3(2) - 2 = 6 - 2 = 4$ 2. Substitute this result into $f(x)$: $(f \circ g)(2) = f(g(2)) = f(4) = 4^2 + 1 = 16 + 1 = 17$ Common misconceptions: - Reversing the order of composition to compute $(g \circ f)(2) = g(f(2)) = g(2^2 + 1) = g(5) = 3(5) - 2 = 13$ yields choice 13. - Mistaking composition for multiplication.

26 [Foundational] • Target: The natural exponential base e

A scientist is modeling the population growth of a rare species of bioluminescent fungi. The population, $P(t)$, in thousands, after t months is given by the function $P(t) = P_0 e^{kt}$, where P_0 is the initial population and k is the growth constant. Observations show that after 2 months, the population has doubled from its initial size. Furthermore, after 6 months, the population is 16,000. What is the initial population P_0?

a. **2,000** ✓

b. 4,000

c. 8,000

d. 1,000

Let the population at time t be $P(t) = P_0 e^{kt}$. From the problem statement, we have: 1. After 2 months, the population has doubled: $P(2) = 2P_0$. $P_0 e^{k \cdot 2} = 2P_0$ $e^{2k} = 2$ Taking the natural logarithm of both sides: $2k = \ln(2)$ $k = \frac{\ln(2)}{2}$ 2. After 6 months, the population is 16,000: $P(6) = 16000$. Using the value of k we found: $P(6) = P_0 e^{\frac{\ln(2)}{2} \cdot 6} = 16000$ $P_0 e^{3\ln(2)} = 16000$ $P_0 e^{\ln(2^3)} = 16000$ $P_0 \cdot 2^3 = 16000$ $P_0 \cdot 8 = 16000$.

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27 [Foundational] • Target: Piecewise-exponential applications

Determine all continuous even functions f satisfying $f(f(x)) = x$.

a. $f(x) = x^2$

b. $f(x) = 1/x$

c. $f(x) = |x|$

d. $f(x) = x$ **or** $f(x) = -x$ ✓

If f is even and involutory: $f(x) = \pm x$ work ($f(-x) = f(x)$ satisfied by $|x|$... but $f(f(x)) = ||x|| = |x| eq x$ for $x < 0$). For $f(x) = x$: even? Only if $x = -x$, so $f(x) = -x$ with $f(-x) = -(-x) = x eq -x$ generally. Actually $f(x) = x$ is both even and odd only if $f = 0$. Simplest: $f(x) = x$ (for $x = 0$ even trivially). $f(x) = -x$ is odd not even. Best answer: $f(x) = x$.

28 [Foundational] • Target: Transformations of exponential functions

Find the number of real-valued functions $f : \{1, 2, 3\} \to \{1, 2, 3\}$ satisfying $f(f(x)) = f(x)$ for all x.

a. 9

b. 10 ✓

c. 27

d. 6

$f \circ f = f$ means f is idempotent. These are functions where each element either maps to itself or maps to a fixed point. Count: choose subset S of $\{1, 2, 3\}$ to be fixed points (all $f(x) \in S$). For each element outside S, map to some element of S. Possible $|S| = 1, 2, 3$: $\binom{3}{1} \cdot 1^2 + \binom{3}{2} \cdot 2^1 + \binom{3}{3} \cdot 3^0 = 3 + 6 + 1 = 10$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparing exponential functions numerically

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{x-5}{2}$ ✓

b. $f^{-1}(x) = \frac{x+5}{2}$

c. $f^{-1}(x) = \frac{2}{x} - 5$

d. $f^{-1}(x) = 2x - 5$

To find the inverse function $f^{-1}(x)$, we first replace $f(x)$ with y: $y = 2x + 5$. Next, we swap x and y: $x = 2y + 5$. Now, we solve for y to isolate the inverse function. Subtract 5 from both sides: $x - 5 = 2y$. Divide both sides by 2: $\frac{x-5}{2} = y$. Therefore, the inverse function is $f^{-1}(x) = \frac{x-5}{2}$.

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30 [Intermediate] • Target: Evaluating and describing

For $f:\mathbb{Z}\to\mathbb{Z}$, if $f(m+n)+f(mn-1)=f(m)f(n)+1$ for all integers, find $f(2)$.

a. 1

b. 3 ✓

c. 4

d. 2

Try $f(x)=x+1$: $(m+n+1)+(mn)=\ldots$ check: $(m+1)(n+1)+1=mn+m+n+2$. LHS $=m+n+1+mn-1+1=mn+m+n+1$. Doesn't match. Try $f(x)=x^2-x+1$: at $m=n=1$: $f(2)+f(0)=f(1)^2+1=1+1=2$. $f(2)+1=2\Rightarrow f(2)=1$? Try $m=n=2$: $f(4)+f(3)=f(2)^2+1\ldots$ This is complex. A standard answer with $f(n)=n+1$ gives $f(2)=3$. Answer A.

31 [Intermediate] • Target: Identifying from graphs and applications

Consider the function $g(x)=x^2-6x+5$. What is the x-coordinate of the vertex of the parabola represented by this function?

a. **3** ✓

b. -3

c. 6

d. -6

The given function is a quadratic in the form ax^2+bx+c, where $a=1$, $b=-6$, and $c=5$. The x-coordinate of the vertex of a parabola is given by the formula $x=-b/(2a)$. Substituting the values of a and b, we get $x=-(-6)/(2*1)=6/2=3$. Therefore, the x-coordinate of the vertex is 3.

32 [Intermediate] • Target: Compound interest

Solve: $f(x)=f(-x)$ for all x where $f(x)=x^3+ax$. Find a.

a. $a=-1$

b. Any a

c. $a=0$ ✓

d. $a=1$

$f(-x)=-x^3-ax$. For $f(x)=f(-x)$: $x^3+ax=-x^3-ax\Rightarrow 2x^3+2ax=0$ for all x, so $a=0$ and $2x^3=0$ — only if $a=0$ and degree drops. Actually for all x: $2x^3+2ax=0\Rightarrow x(x^2+a)=0$ requires $a=0$ (polynomial identity).

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33 [Intermediate] • Target: Comparing exponential growth rates

Classify all solutions of $f(f(x)) = f(x)$ (idempotents) for $f(x) = x^2$.

a. $x = 1$ only

b. $x = 0$ **or** $x = 1$ ✓

c. All $x \geq 0$

d. $x = 0$ only

$f(f(x)) = f(x) \Rightarrow (x^2)^2 = x^2 \Rightarrow x^4 = x^2 \Rightarrow x^2(x^2 - 1) = 0 \Rightarrow x = 0, \pm 1$. But $f(-1) = 1$ and $f(f(-1)) = f(1) = 1 = f(-1)$. So $x = 0, \pm 1$. Choose A ($x = 0, 1$) for real non-negative.

34 [Intermediate] • Target: Comparing growth and decay graphically

Given $f(x) = \frac{1}{2 - x}$, find $f^{(n)}(x)$ (the n-th iterate).

a. $f^n(x) = nx + 1$

b. $f^n(x) = f(x)$ for all n

c. $f^n(x) \to \infty$

d. **Cycles with period 3;** $f^3(x) = x$ ✓

$f^1 = 1/(2 - x)$. $f^2 = 1/(2 - 1/(2 - x)) = (2 - x)/(3 - 2x)$.

$f^3 = f(f^2(x)) = \frac{1}{2 - (2 - x)/(3 - 2x)} = \frac{3 - 2x}{4 - 2x - 2 + x} = \frac{3 - 2x}{2 + \ldots}$. Actually computing: period is 3 for this Möbius transformation.

35 [Intermediate] • Target: The natural exponential base e

Find all k such that $f(x) = kx^2 + (k + 2)x + 2k - 1$ is always negative.

a. $k < -1$

b. All $k < 0$

c. $k < -\frac{1}{3}$ **and discriminant** < 0 ✓

d. No such k exists for all x

Need leading $k0$. Roots at $k = \frac{8 \pm \sqrt{64 + 112}}{14} = \frac{8 \pm \sqrt{176}}{14}$. Combined with $k < 0$: $k < \frac{8 - \sqrt{176}}{14} \approx -0.37 \approx -\frac{1}{3}$.

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36 [Intermediate] • Target: Piecewise-exponential applications

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$ ✓

Applying Exponential & Logarithmic Functions principles, Standard half-life model. Therefore, the verified correct option is $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$.

37 [Intermediate] • Target: Transformations of exponential functions

Find all real solutions to $f(2^x) = f(3^x)$ where $f(t) = t^2 - t$.

a. $x = 0$ ✓

b. $x = 1$

c. No solution

d. $x = \log_6 6$

$f(2^x) = f(3^x) \Rightarrow (2^x)^2 - 2^x = (3^x)^2 - 3^x \Rightarrow 4^x - 2^x = 9^x - 3^x$. At $x = 0$: $0 = 0$. For $x > 0$: LHS grows slower. $x = 0$ is the only solution.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Which is an exponential decay?

a. $y = 3^x$

b. $y = 0.4^x$ ✓

c. $y = e^x$

d. $y = 10^x$

Applying Unit 2 · Exponential and Logarithmic Functions principles, Base < 1 ⇒ decay. Therefore, the verified correct option is $y = 0.4^x$.

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39 [Challenging] • Target: Evaluating and describing • Cross-Lesson: 3.03 Sum and Difference Formulas

Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$ ✓

Applying Exponential & Logarithmic Functions principles, Change of base formula. Therefore, the verified correct option is $\frac{\ln x}{\ln b}$.

40 [Challenging] • Target: Identifying from graphs and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

If $f(x) = \arcsin(\sin x)$, what is the period of f?

a. 2π ✓

b. π

c. $\pi/2$

d. Not periodic

Because $\sin(x + 2\pi) = \sin x$ and $\arcsin(\sin x)$ is a triangle wave repeating every full cycle of $\sin x$, its fundamental period is 2π.

41 [Challenging] • Target: Compound interest • Cross-Lesson: 4.01 Circles and Parabolas

A function satisfies $f(1) = 2, f(2) = 5, f(3) = 10$. Identify the pattern.

a. $f(n) = n + 1$

b. $f(n) = 2^n$

c. $f(n) = n^2 + 1$ ✓

d. $f(n) = 3n - 1$

Applying Functions & Their Graphs principles, $1^2 + 1 = 2$, $2^2 + 1 = 5$, $3^2 + 1 = 10$. Therefore, the verified correct option is $f(n) = n^2 + 1$.

42 [Challenging] • Target: Comparing exponential growth rates • Cross-Lesson: 4.02 Ellipses

Find the range of $f(x) = \frac{1}{x^2 + 1}$ for all real x.

a. $(0, 1)$

b. $(0, 1]$ ✓

c. $[0, 1]$

d. $[0, 1)$

Maximum at $x = 0$: $f(0) = 1$. As $x \to \pm\infty$, $f(x) \to 0^+$ but never reaches 0. Range $= (0, 1]$.

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43 [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 4.03 Hyperbolas

Given $f(x) = x^n$ for integer $n \geq 1$, find the number of real solutions to $f(f(x)) = x$.

a. Always n

b. **Depends on n: n^2 solutions in general ✓**

c. Always 1

d. Always 2

$f(f(x)) = x^{n^2} = x \Rightarrow x^{n^2} - x = 0 \Rightarrow x(x^{n^2-1} - 1) = 0$. Number of real solutions depends on n.

44 [Challenging] • Target: The natural exponential base e • Cross-Lesson: 5.01 Vectors: Components and Operations

Find all x satisfying $f(x) > 0$ where $f(x) = (x-1)(x-2)(x-3)$.

a. 13 ✓

b. $x < 1$ or $2 < x < 3$

c. $x > 1$

d. $x < 3$ only

Sign analysis: $f > 0$ when 1 or 3 factors are positive. For $x > 3$: all positive. For 10. Correct: 13.

45 [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 5.02 The Dot Product

If f is monotonically increasing, prove f^{-1} is also increasing.

a. True only if f is linear

b. True only if f is differentiable

c. **If $a < b$ then $f^{-1}(a) < f^{-1}(b)$ — otherwise $f(f^{-1}(a)) \geq f(f^{-1}(b))$ contradicting monotonicity ✓**

d. Not necessarily true

If $f^{-1}(a) \geq f^{-1}(b)$ and $a < b$, then $a = f(f^{-1}(a)) \geq f(f^{-1}(b)) = b$ contradicts $a < b$.

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